

SYNOPSIS

At 0607 Universal Co-ordinated Time (UTC) on 14 June 1998, the UK registered fishing vessel *Silvery Sea* collided with the German registered container ship *Merkur* about 35 miles west of Esbjerg, Denmark. This caused *Silvery Sea* to founder with the loss of her five-man crew. The Department of the Environment, Transport and the Regions' (DETR) duty officer notified the Marine Accident Investigation Branch (MAIB) of the accident at 0912 that day. Captain P Kavanagh carried out the investigation.

Silvery Sea was on passage from the fishing grounds near the entrance to the Firth of Forth on a true course of 098° and making a speed of about 9.3 knots. She was fully loaded with her catch of sand eels and making for Esbjerg on the west coast of Denmark to land it.

Merkur was on passage from Hamburg to Gothenburg, partly loaded with containers. She was making good a course of 340° and a speed of 15 knots. The chief officer and a seaman were on watch at the time of the incident. The weather was fine with good visibility and it was daylight. There was no other traffic involved in the incident.

Shortly after 0530, *Merkur*'s chief officer saw an echo on the edge of his radar screen. It was on his port bow at a range of about 7 miles. As the bearing of the echo did not change, he assessed that a risk of collision existed. Using his binoculars, he identified the echo as a fully-laden fishing vessel on an easterly heading, probably making for Esbjerg. She was not engaged in fishing. A crossing situation existed, and by the *International Regulations for Preventing Collisions at Sea (Collision Regulations)* it was the fishing vessel's responsibility to keep out of the way of the vessel on her starboard bow, the German container ship. The chief officer, in the stand-on vessel, maintained his course and speed.

There is some doubt as to whether or not the chief officer left the bridge before the collision. However, the evidence indicates that at 0600, he plotted *Merkur*'s position and entered it into the deck logbook. He then returned to the navigation console and saw that the fishing vessel had taken no avoiding action. Realising there was a risk of an imminent collision, he sounded a warning signal on the whistle. The fishing vessel did not respond, and although *Merkur*'s chief officer altered course to starboard, it was too late and the two vessels collided.

The fishing vessel sank very quickly. *Merkur*'s chief officer sounded the general alarm, called the master and made a distress call. It was received by Lyngby Radio and despite an extensive search by a number of vessels and a helicopter, no survivors were found.

Later on, Danish divers retrieved the bodies of all five fishermen from the wreck.

The cause of the collision was twofold. *Silvery Sea* did not keep out of the way of *Merkur*, and *Merkur*'s chief officer did not take avoiding action sufficiently early to prevent a collision.

Silvery Sea foundered because she was holed forward and pushed down in the collision. Water was able to progressively flood the full length of the vessel through open doors in the transverse bulkheads.

Recommendations are made with regard to collision bulkheads in fishing vessels, to the researching of possible increased deadweight of RSW tanks due to the settling and compression of sand eels, and to the review of criteria for commissioning postmortem examinations following an accident.

SECTION 1 - FACTUAL INFORMATION

1.1 NARRATIVE

All times are UTC except as otherwise stated. Local time is UTC + 2. All courses are true.

1.1.1 Events leading up to the collision

Silvery Sea

During June, there is a short season for catching sand eels on the fishing grounds of Wee Bankie, near the entrance to the Firth of Forth. Denmark is one country which processes sand eels into fishmeal/fertiliser.

The fishing vessel *Silvery Sea* was engaged in sand eel fishing and had landed her first catch of the season at Esbjerg on 6 June 1998. She then returned to Wee Bankie to make her second catch and on 12 June, while in contact with his home port of Mallaig, her skipper said that despite a poor start, the day's fishing had been good. He added that he was fully loaded and would be returning to Esbjerg.

In a fax to his wife on 13 June, the skipper said *Silvery Sea* was making 9.3 knots. The course from Wee Bankie to where the accident took place was about 098° and the distance about 317 miles. Had she maintained a speed of 9.3 knots throughout she would have been underway for about 34 hours, and would therefore probably have set sail from the fishing grounds at about 2000 on 12 June.

In a fax to his Danish agent, the skipper gave his estimated time of arrival (ETA) in Esbjerg as 1400 (local time) on 14 June, and said he had approximately 530 tonnes of sand eels on board

During passages across the North Sea, each member of the crew normally expected to take two-hour navigational watches in turn. Watches were allocated randomly.

It is not known who was on watch when the collision occurred at around 0607, and there is no information available to indicate what did, or did not happen on board *Silvery Sea* in the 30 minutes or so that preceded it. As visibility was good, it is assumed that the German cargo ship would have been clearly visible on *Silvery Sea*'s starboard bow during the minutes leading up to the collision.

Merkur

Merkur left Hamburg for Gothenburg at 1900 on 13 June. She was loaded with containers in her holds and with one layer on top of No 2 and No 3 hatches (see **Diagram 1**). One container was on a second layer directly in front of the accommodation: her maximum load of containers could reach up to four high on the hatches.

The master took the first navigational watch and was relieved at 0006 the next day by the second officer. *Merkur* had, by then, left the Elbe and was on her first course of 342°, to the first waypoint 2.5 miles due west of the Horn Rev west cardinal buoy and some 35 miles west-north-west of Esbjerg.

The chief officer relieved the second officer at 0300, and was joined an hour later by a new watchman who, after drinking a cup of coffee, was employed cleaning the starboard bridge wing.

The chief officer plotted *Merkur*'s position every half hour using the global positioning system (GPS), and entered the hourly positions in the logbook. The radar was on the 6-mile range scale but was off-centred giving a 9-mile range ahead. The bridge watch alarm system was set to sound at 12-minute intervals.

Between 0500 and 0600 *Merkur* made good a course of 340° and a speed of 15 knots. There were no other vessels near by. Between 0500 and 0515, the watchman stopped cleaning the bridge wing and went below for breakfast. This normally took him about 30 minutes. When he returned he resumed his cleaning.

Shortly after plotting the 0530 position, the chief officer saw an echo on the edge of the radar screen and on the port bow at about 7 miles range. Using a pair of binoculars he identified it as a fishing vessel on an easterly heading. She was not engaged in fishing as she was not showing fishing signals and had no fishing gear deployed. Because her amidships freeboard was small he estimated she was fully loaded.

At about 0545, using the electronic bearing marker (EBM) on the radar, the chief officer noted the fishing vessel was on a steady bearing. Because she was on his port bow and therefore the give-way vessel in a crossing situation, he assumed she would obey the *Collision Regulations* and keep out of *Merkur*'s way. He therefore maintained his course and speed.

The chief officer told the watchman to keep a good lookout on the starboard side for fishing vessels outbound from Esbjerg.

There is some doubt as to whether or not the chief officer left the bridge before the collision. However, the evidence indicates that after plotting the 0600 position he looked up from the chart table and saw that the fishing vessel had not given way and, with the range now at about 8 cables was, seemingly, maintaining her original course and speed. The chief officer sounded a warning signal on the whistle. Seeing there was no response from the fishing vessel and that a dangerous close-quarters situation was developing, he put the rudder hard-to-starboard about 1 minute after sounding the whistle. As the ship's head began to turn, the fishing vessel collided with *Merkur* just abaft her port bow. The impact forced the fishing vessel's head to port, and her starboard side made further contact with *Merkur*'s port quarter.

The chief officer sounded the general alarm, called the master and transmitted a "Mayday" on very high frequency (VHF) radio channel 16. Lyngby Radio responded, and the radio operator asked for the ship's position, which was given as Latitude

about 0200 the following morning, the level of water in the holds had stabilised. *Merkur* then made for Esbjerg, where she berthed at 1124 on 15 June.

1.2 ENVIRONMENTAL CONDITIONS

At the time of the collision, the wind was south-east force 5 with good visibility. The seas were slight and there was no swell.

The tide was setting in a north-westerly direction at 0.5 knot.

The altitude of the sun was just over 23° on a bearing of about 082°.

The surface sea water temperature was about 11° C.

1.3 SILVERY SEA

1.3.1 Background

Silvery Sea was the largest fishing vessel to sail out of Mallaig and all but one of her crew came from that area. The skipper was held in high regard and imposed exacting standards of discipline, seamanship and working practices on his crew. *Silvery Sea* was known as one of the better-operated vessels in the Scottish fleet.

She was primarily a purse seine netter, but was also equipped with trawl nets.

A current United Kingdom Fishing Vessel Certificate as required by *The Fishing Vessels (Safety Provisions) Rules 1975*, was held by *Silvery Sea* and had been issued on 30 January 1998.

1.3.2 Vessel description

Silvery Sea was built with an overall length of 27.1m and had three refrigerated saltwater (RSW) tanks. Forward of the tanks were a dry fish hold and a conventional whaleback. The accommodation consisted of three decks: the wheelhouse deck, a galley/bathroom/mess deck and two six-man berths below. The skipper's cabin was on the after port side of the wheelhouse.

In 1985, *Silvery Sea* was lengthened with a 10m full-body section (comprising three extra RSW tanks) being inserted amidships. An enclosed steel shelter deck was constructed from the top of the whaleback deck to the forward bulkhead of the wheelhouse. The old RSW tanks were extended upwards to the shelter deck level and fitted with hatches. The accommodation was heightened by 2m by inserting an extra deck between the wheelhouse and the mess deck which then became a recreation room (see Diagram 2).

The shelter space formed part of the buoyancy and was included in the stability calculations. For the converted vessel, this extra buoyancy was necessary to pass the

requirements for stability. For stability purposes, the uppermost deck therefore, moved from the main deck to the shelter deck.

When built the vessel had a collision bulkhead on frame 46, which due to the lengthening became frame 71. Originally, the collision bulkhead extended to the uppermost continuous deck, which was the main deck. No collision bulkhead exemption was necessary, or granted, for the original arrangement.

Rule 2 of *The Fishing Vessels (Safety Provisions) Rules 1975* requires that a collision bulkhead is fitted, but makes no detailed requirements. Had the cargo ship construction regulations been followed, such a bulkhead would have been required to extend up to the shelter deck with no openings.

After *Silvery Sea* was lengthened, the SGO (now the MCA) granted an exemption from the requirement to have a collision bulkhead. The reason for this exemption is unknown.

In the gutting room there was a hatchcover to the dry fish hold. At the forward end of the gutting room a weathertight door, secured by two dogs and on the starboard side at frame 74, gave access to the forward store (**see Photograph 1**). At the after end were two passageways, which ran aft along both outboard sides of the RSW tanks. The passageways did not have weathertight doors at the forward ends, but they were fitted aft. These gave access to the RSW machinery space. In the transverse bulkhead at the after end of the RSW machinery space, there was one weathertight door, which provided access to the accommodation and engine room spaces. There was also a door on the port side of the bulkhead at the forward end of the gutting room.

A net drum was added at the stern in 1986.

In May 1987, following an engine failure, *Silvery Sea* was re-engined with a more powerful one but, due to the constraints of the propeller blade size, it was restricted to give the same power output as the original.

She had three sonars: one low frequency/long range set, one high frequency/short range set and a back-up set. For navigation she had an Admiralty raster chart service (ARCS) electronic chart system, which had a GPS input, and two radars. She had three VHF radio sets: two medium frequency radiotelephone sets, and a SATCOM type 'C' laptop computer for sending faxes.

There were two bridge chairs, one in the centre and the other on the starboard side of the wheelhouse. The sonars, the electronic chart system, automatic helm, engine controls, watch alarm and other ancillary equipment were arranged, close at hand, around the starboard chair. In front of the centre chair were secondary engine controls.

A watch alarm was fitted in the wheelhouse and located to the right of the starboard wheelhouse chair. The watch alarm was automatically activated when the autopilot was engaged (**see Photograph 2**). It could not be heard in the lower deck cabins because of the ambient noise from the propeller and onboard machinery. When in

SECTION 2 - ANALYSIS

2.1 AIM

The purpose of the analysis is to determine the cause, contributory causes and circumstances of the accident as a basis for making recommendations, if any, with the aim of preventing similar accidents occurring again.

This section will firstly examine how these two vessels collided in good visibility and weather, and secondly, why *Silvery Sea* foundered so quickly that none of the crew was able to escape

2.2 THE COLLISION

2.2.1 *Silvery Sea*

Silvery Sea was not engaged in fishing within the meaning of Rule 3(d) of the *International Regulations for Preventing Collisions at Sea* (Collision Regulations), and her manoeuvrability was not restricted. She was on passage from the fishing grounds off the Firth of Forth to her landing port of Esbjerg, making a course of about 098° and a speed of about 9.3 knots (see Diagram 3 overleaf). The autopilot dial indicated the course being steered was between 100° and 103°.

Merkur was about 38° or 3½ points on *Silvery Sea*'s starboard bow. As events evolved, and from the chief officer's evidence, the bearings of each vessel from one another would not have changed appreciably, indicating they were on a collision course. This was a crossing situation as defined by Rule 15 of the Collision Regulations. The rule states that "*the vessel which has the other vessel on her own starboard side shall keep out of the way and shall, if the circumstances of the case permit, avoid crossing ahead of the other vessel.*" In practical terms this means that the give-way vessel can either reduce speed and allow the stand-on vessel to cross ahead, or, more usually, alter course to starboard and pass round the stern of the other vessel. Rule 16, of the Collision Regulations, states that "*the give-way vessel should take, so far as possible, early and substantial action to keep well clear.*" *Silvery Sea* had plenty of sea room in which to manoeuvre.

The evidence from the underwater survey indicates that when she collided with *Merkur*, *Silvery Sea* was on a steady course, and on full economical speed.

It is concluded that whoever was on watch on *Silvery Sea* was not, in the period leading up to the collision, keeping a proper lookout in accordance with Rule 5 of the Collision Regulations, and had failed to take any action to avoid collision. There is no obvious reason why.

The question arises as to how a well-run vessel, fitted with a working watch alarm, could collide with another ship without, apparently, any action being taken to prevent it.

Several theories have been examined. Fatigue is a common cause of accidents in fishing vessels when either fishing or on passage to land. *Silvery Sea*'s crew should have been well rested in the 34 hours since they had left the fishing grounds, so there was no apparent reason why any of them would have been suffering from undue tiredness or sleep deprivation. Furthermore, as each member of the crew normally kept only one two-hour watch in every ten, fatigue is unlikely to have been the reason why nobody was keeping a proper lookout. Even if someone had been in the wheelhouse and asleep, it is unlikely he would have slept through the loud watch alarm.

The person on watch may have left the bridge for some reason and stayed below for several minutes. Had he done so the 110db watch alarm klaxon would have sounded after 4 minutes. This is not only audible on the wheelhouse, but also in the recreation room and galley. The crew members, who were probably on the lower decks, would not have heard it due to the noise of the propeller, ventilation and engine. Had the skipper been in his cabin and conscious he would have heard the klaxon. Although it is assumed the watch alarm was functioning normally, it is possible it was not.

Whatever the reason, the watch alarm failed to alert the crew to the fact that nobody was maintaining a watch in the period before the collision.

There is no evidence to show that the sound signals from *Merkur*, when the range had closed to about 8 cables, did anything to alert *Silvery Sea* to the risk of collision.

It is also feasible the watchkeeper had been taken ill in the wheelhouse, and either remained there or had left the bridge unattended to recover. In either situation the alarm would have sounded, and the same considerations would have applied.

Because of the random nature of the roster system, it is impossible to determine who was, or should have been, on watch just before the collision.

The investigation also considered whether there was any physical or other reason to prevent the keeping of a good lookout. The visibility was good, there is no evidence to indicate there was any other shipping to distract the watchkeeper's attention, and so far as it is possible to judge, the bridge equipment was in working order. There were no adverse environmental conditions, such as glare from the early morning sun reflecting on the surface of the sea. The difference in bearing between the sun as it was rising on the port bow and the bearing of *Merkur* on the starboard bow was about 55°. Any glare would have had minimal effect.

The positions in which the bodies were found give no indication as to whether the bridge was manned at the time of the collision or where people were when it happened. There would have been an enormous shudder at the moment of impact, which would have woken anyone who was asleep. Even allowing for momentary disorientation, it is probable that some or all of the crew attempted to move from wherever they were to either find out what had happened, or to escape once it was known how serious the situation was. It is also likely that the inrush of water swept them to where they were found.

Mr Dyer's body was found on the lower deck. His facial injuries indicated that he might have been knocked unconscious, either from the sudden jolt of the collision or from the large ingress of water. He did not have suffusion of the face, which is the usual sign of drowning.

Mr Tait was found with a lifejacket at the foot of the stairs to the wheelhouse. He was known to keep his lifejacket near his bunk, but lifejackets were also stored in the wheelhouse.

It is impossible to reconstruct precisely what happened during the short time between collision and sinking. The only certainty about the entire incident is that *Silvery Sea* sank very soon after the collision.

The skipper's body was found at the bottom of the wheelhouse stairs at the entrance to the recreation room. Like Mr Dyer, he did not have suffusion of the face. The pathologist gave a number of reasons for the possible causes of the skipper's death, but none were conclusive. No postmortem was carried out and without this crucial input, it has not proved possible to establish the cause of death.

The Danish Police authorities carried out formal identification procedures of all the bodies, giving the cause of death of each fisherman as drowning. These conclusions were made without the benefit of postmortem examinations. The Scottish authorities did not carry out any autopsies on the deceased fishermen and, because the accident happened outside Scottish territorial waters, the *Fatal Accident and Sudden Death Inquiry (Scotland) Act* of 1976 does not provide for a Fatal Accident Inquiry to be convened.

The lack of any postmortems on the deceased prevented the investigation from establishing whether any of them died from causes other than drowning.

The investigation has been unable to establish why a proper lookout was not kept onboard *Silvery Sea* in the minutes before, and at the time of, the collision with *Merkur*.

Silvery Sea's wheelhouse battery-powered clock stopped at 0709 (UTC+1), which probably occurred when it was immersed in water as the vessel foundered. The vessel foundered several minutes after the collision, which is estimated as having occurred at 0607 (UTC). As the distress message from *Merkur* was received at 0612, the estimated collision time is a reasonable deduction.

2.2.2 *Merkur*

On board *Merkur*, shortly after 0530, the chief officer first saw the echo of a fishing vessel on his port bow on the edge of his off-centred radar screen and at a range of 7 miles. However, with the radar on the 6-mile range scale, the off-centred setting extended the range ahead to 9 miles. It has been calculated that the actual range of the echo at the time of sighting would have been about 7.6 miles. On reconstruction, the first sighting on the radar would therefore have been at about 0546, so the echo was first seen on radar some 21 minutes before the collision.

He knew from the EBM that the vessel was on a steady bearing and therefore on a collision course. The evidence is inconclusive about how carefully he monitored the approach of the fishing vessel, but all the indications show that he was in no doubt that a risk of collision existed from when he first started to monitor it. Furthermore, he recognised *Silvery Sea* was not engaged in fishing. He knew he had a crossing situation and the fishing vessel was, by Rule 15 of the Collision Regulations, the give-way vessel.

Merkur's chief officer soon assessed he was the stand-on vessel and therefore obliged to observe Rule 17 of the Collision Regulations. The rule includes the following:

- (a) (i) *Where one of two vessels is to keep out of the way the other shall keep her course and speed.*
- (ii) *The latter vessel may however take action to avoid collision by her manoeuvre alone, as soon as it becomes apparent to her that the vessel required to keep out of the way is not taking appropriate action in compliance with these rules.*
- (b) *When, from any cause, the vessel required to keep her course and speed finds herself so close that collision cannot be avoided by the action of the give-way alone, she shall take such action as will best aid to avoid collision.*

It is evident from the available information that, from the time it was ascertained that a risk of collision existed until apparently the range had closed to about 8 cables, very little monitoring of the fishing vessel was being undertaken. There is some suggestion to indicate the chief officer may not even have been on the bridge for some of this time, and the watchman had been directed to keep a lookout for vessels approaching from the opposite, starboard, side. During the few minutes preceding the collision the evidence suggests that the chief officer was occupied with plotting and logging the ship's position. Whatever actually happened, the indications are that it was not until the range had closed to about 8 cables that he realised the fishing vessel was not taking appropriate action to keep out of the way.

Under Rule 34 (d) of the Collision Regulations, the chief officer gave a warning signal on the ship's whistle to show his doubt as to whether sufficient action was being taken by the fishing vessel to avoid collision.

Rule 34(d). *"When vessels in sight of one another are approaching each other and from any cause either vessel fails to understand the intentions or actions of the other, or is in doubt whether sufficient action is being taken by the other to avoid collision, the vessel in doubt shall immediately indicate such doubt by giving at least five short and rapid blasts on the whistle."*

The chief officer made the sound signal after he had plotted and logged the 0600 position and when he estimated that the fishing vessel was about 8 cables away.

The range was then very close indeed. From the reconstruction, a distance apart of only 8 cables equates to about 2 minutes steaming time to the point of impact. The chief officer might have chosen to use the VHF radio to call the fishing vessel but did

not do so because he could not see her name. He thought that if he called the fishing vessel without giving her name, another vessel might have answered, which would have caused confusion and wasted time in avoiding her. There is no disagreement with this decision, but he should have been alert to his responsibility under Rule 2(a) of the Collision Regulations, which states:

“Nothing in these Rules shall exonerate any vessel, or the owner, master or crew thereof, from the consequences of any neglect to comply with these Rules or of the neglect of any precaution which may be required by the ordinary practice of seamen, or by the special circumstances of the case.”

Although the chief officer chose not to take action in accordance with Rule 17(a)(ii), he effectively had a responsibility under Rule 2(a) to take avoiding action before a point had been reached where a collision could not be avoided by the combined actions of both vessels. If he had delayed taking action until a collision could not be avoided by the action of the other vessel alone, in accordance with Rule 17(b), a collision would probably still have resulted.

Merkur had plenty of sea room in which to manoeuvre, but not the luxury of time. The chief officer waited a further minute before altering course to hard-to-starboard. The ship's data shows that, with one steering motor, it takes 32 seconds for the rudder to be moved hard over to 45°, and a further minute for the ship to turn through 90°. The chief officer should have sounded the whistle much sooner and, having seen no reaction from the fishing vessel, altered course much earlier. His actions were too late and, therefore, inadequate to avoid collision.

When the master arrived on the bridge, he could see from the wake that his ship was turning to starboard and that the still floating fishing vessel was on the starboard quarter.

The damage to *Silvery Sea* shows a bias to the port bow while the scrape marks on *Merkur* indicate that the fishing vessel impacted at an angle in excess of 90°. If *Merkur* had altered course to starboard earlier, the angle between them would have been less. Therefore, *Merkur* must have only just started to turn when the collision occurred.

The question arises as to why an experienced officer left it so late before taking any action to avoid a collision. There are two possible reasons:

Firstly, he might not have been on the bridge in the minutes preceding the collision. In their testimonies, the police officers stated, separately, that different people had told them that there was nobody on the bridge. Despite lengthy and detailed questioning by the German Board of Inquiry panel, the policemen stood by their testimony (see **Section 1.5**). The inquiry concluded that the chief officer temporarily left the bridge. If this was so, why did he leave the bridge when he knew that a fishing vessel was approaching on a collision course?

The chief officer said that he had not left the bridge, and the watchman corroborated this. It can only be speculated as to why he would want to leave the bridge, and this

at about 0545, after which the chief officer could have left him alone on watch while he went for a quick breakfast before returning to take the 0600 position. Normally he would have been relieved at 0600 but, because he was not being relieved until 0700 that day, he might have missed breakfast. He may, alternatively, have left the bridge to visit the toilet, but he denied this at the Inquiry.

The second reason follows from the chief officer's own explanation as to why he did not alter course until the fishing vessel was so close. He said he had assumed the fishing vessel would alter course and saw no reason why she would not do so. With his experience of coastal trading, and under the circumstances in which he found himself, the tolerance threshold for the distance he expected fishing vessels to alter course for his ship was much less than for merchant ships. He knew that fishing vessels could, when on passage, alter course quickly and do so at close range. He made this assumption on the basis that the fishing vessel watchkeeper would be keeping a proper lookout. It is a dangerous assumption and no officer of the watch should ever make such assumptions. He should always be alert to the possibility that such a watch might not be kept, should keep watching the give-way vessel with care and, if necessary, judge the timing of his avoiding actions accordingly.

There is a conflict of evidence between that of the policemen and that of the master, chief officer and the watchman. It is impossible to draw a definite conclusion as whether or not the chief officer had left the bridge at any time, or whether this was even a contributory cause of the collision. It was unfortunate that MAIB inspectors were unable to interview the crew of *Merkur* after the incident.

2.3 THE FOUNDERING OF *SILVERY SEA*

Why did this large well-founded but heavily laden fishing vessel sink so quickly that none of the crew were able to escape? The answer to this question lies mainly in two aspects - the loaded condition of the vessel and her internal watertight integrity.

2.3.1 The loaded condition

The stability book showed that the total capacity of the RSW tanks was 433.1 tonnes. On examination it was found that this capacity was calculated with a sea water density (or specific gravity - SG) of 0.976. This was wrong and should have been 1.025*, changing the capacity to 454.84 tonnes. Condition 4 of the stability book relates to departure from the fishing grounds with 100% full RSW tanks, 50% fuel and 50% fresh water. The waterline for this condition has been drawn on **Diagram 4**.

(* Note: *The density of sand eels in RSW tanks has been taken to be the same as sea water.*)

The MAIB re-defined the hull of *Silvery Sea* to undertake trim and stability calculations, and confirmed the assessment given above.

In the skipper's fax to his agent (see **Section 1.1.1**), he said that there were 530 tonnes of (sand eels) on board. It is generally agreed that this was probably an over-

SECTION 3 - CONCLUSIONS

3.1 FINDINGS

3.1.1 General

1. A collision occurred between the UK fishing vessel *Silvery Sea* and the German container ship *Merkur* at about 0607 UTC on 14 June 1998, in a position approximately 35 miles west of Esbjerg, Denmark. [1.1.1]
2. No environmental conditions contributed to the accident. [2.2.1]
3. There was no other traffic in the area, which had a bearing on the collision. [1.1.1]
4. Both vessels had plenty of sea room in which to manoeuvre. [2.2.1, 2.2.2]
5. Postmortem examinations on the deceased fishermen, by either the Danish or Scottish authorities, would have been of assistance in investigating the cause of the collision. [2.2.1]
6. There was no known technical or material deficiency on either vessel that contributed to the cause of the collision. [2.2.1, 2.2.2]

3.1.2 *Silvery Sea*

1. The fishing vessel was on passage from the fishing grounds of Wee Bankie, near the entrance to the Firth of Forth, to her landing port of Esbjerg. [1.1.1]
2. The skipper sent a fax to his agent in Esbjerg stating that he had 530 tonnes of fish on board. This would have been an overloaded condition. [1.3.4]
3. On a previous occasion, *Silvery Sea* had landed a maximum load of fish at Esbjerg of 510 tonnes, which was also an overloaded condition. [1.3.4]
4. There is uncertainty over the SG of compressed sand eels within RSW tanks. [2.3.1]
5. From the fishing grounds to the collision position, the course was 098° and she had reported that she had been making 9.3 knots. [1.1.1]
6. The fishing vessel had been on passage for about 34 hours up to the time of the collision. [1.1.1]
7. Having taken a roster system of two-hour navigational watches for each crew member since leaving the fishing grounds, it is probable that the five fishermen on board would not have been suffering from fatigue. [1.1.1]

8. Due to the random nature of the roster system of watches it is not known who was on watch at the time of the immediate events leading up to the accident. [1.1.1]
9. There was a watch alarm system, which was activated when the autopilot was engaged. [1.3.2]
10. The loud klaxon for the watch alarm sounded only in the wheelhouse but was also audible in the skipper's cabin, the recreation room and galley. It could not be heard in the crew's cabins. [1.3.2]
11. The fishing vessel foundered so quickly that all five crew members on board were unable to escape from within the vessel, and lost their lives. [1.1.2, 1.3.5]
12. Both of the liferafts were released automatically from the sunken vessel, but one failed to inflate although both were serviced and up to date. [1.1.2]
13. The EPIRB was automatically released and the Danish authorities received its alert signal at 0622. [1.1.2]
14. Two days after the accident, Danish divers went to the wreck and found that except for the starboard wheelhouse door, all the outside weathertight doors to the accommodation were closed, and all hatches on deck, except one, were closed. [1.3.5]
15. The divers found that the major damage to the vessel was confined to the stem and there was a large opening in the bow. [1.3.5, 1.3.8]
16. The divers recovered the bodies of the skipper and an engineer from the first deck below the wheelhouse, and of two deckhands, from outside the cabins on a deck further down in the accommodation. [1.3.5]
17. The divers returned to the wreck to survey the wheelhouse and found that the engine controls were on full economic speed ahead and that the automatic pilot was engaged on a heading of between 100° and 103°. [1.3.5]
18. The divers found in the forward space that the starboard door at frame 74 between the gutting room and the forward storeroom was open and the hatch to the dry fish hold had been pushed up. [1.3.5]
19. The divers found the body of the fifth fisherman on the second deck down from the wheelhouse. [1.3.5]
20. The manning scale was short of a chief engineer, but this was irrelevant to the accident. [1.3.2]

3.1.3 *Merkur*

1. The container ship was on passage from Hamburg to Gothenburg steering 342° but making good a course of 340° over the ground, with a speed of 15 knots. [1.1.1]
2. The chief officer took charge of the navigational watch at 0300 and a new watchman was posted at 0400. [1.1.1]
3. The chief officer had had an adequate rest period before taking the watch, and would probably not have been suffering from fatigue. [2.2.1]
4. There is no evidence that the chief officer was under the influence of drugs or alcohol. [1.4.2]
5. The chief officer was plotting the position of the ship every 30 minutes, and entered it in the deck logbook every hour. [1.1.1]
6. The bridge watch alarm was set at 12-minute intervals. [1.1.1]
7. The radar was on the 6-mile range scale but was off-centred to give a 9-mile view ahead. [1.1.1]
8. The chief officer observed an echo on the radar screen on his port bow at a range of about 7 miles. [1.1.1]
9. By using the electronic bearing marker he found that the bearing of the echo was not appreciably changing. [1.1.1]
10. He saw that the echo was that of a fully laden fishing vessel, on passage and on an easterly course heading for Esbjerg. [1.1.1]
11. He determined that this was a crossing situation, and that the fishing vessel was the give-way vessel. He maintained his course and speed as the stand-on vessel. [1.1.1, 2.2.2]
12. There is some doubt as to whether or not the chief officer left the bridge before the collision. However, the evidence indicates that at 0600 he plotted the ship's position and entered it into the logbook. [1.1.1]
13. When the fishing vessel was 8 cables away, the chief officer saw she was still on her original course and speed. [1.1.1]
14. He sounded a warning signal to the fishing vessel to which he received no response. [1.1.1]
15. The chief officer then put the helm to hard-to-starboard but his action was too late and the fishing vessel ran stem-on into *Merkur*'s port bow. [1.1.1]

16. *Merkur* launched her lifeboat to search for survivors. [1.1.1]
17. *Merkur* began to take water into her holds and she sought Esbjerg as a port of refuge. [1.1.2]

3.2 CAUSES

3.2.1 The collision

1. *Silvery Sea* did not meet her obligation under the Collision Regulations to keep out of the way of *Merkur*. [2.2.1]
2. The chief officer of *Merkur* did not take appropriate action early enough to best avoid a collision when it became apparent to him that *Silvery Sea*'s action alone would not avoid a collision. [2.2.2]

3.2.2 The foundering

The large ingress of water through the damage in *Silvery Sea*'s bow was able to flow from forward to aft through open doors in transverse bulkheads, thereby depleting her reserve buoyancy and causing the vessel to sink rapidly. [2.3]

3.3 CONTRIBUTORY CAUSES

1. The reason why whoever was on watch in *Silvery Sea* failed to take action to avoid a collision cannot be established because all crew members lost their lives. [2.2.1]
2. It can only be assumed that the watchkeeper on *Silvery Sea* had become incapacitated, or left the wheelhouse and allowed the vessel to continue in automatic helm. [2.2.1]
3. The probable failure of the watch alarm to alert the crew that nobody had cancelled it and, by inference, that there was no-one keeping an effective watch. [2.2.1]
4. The chief officer on *Merkur* made a wrong assumption that *Silvery Sea* was keeping a proper lookout and that she would keep out of the way of *Merkur*, and so did not take avoiding action until it was too late. [2.2.2]
5. Had he used the radar to determine the time of nearest approach of the fishing vessel, the chief officer might not have taken the 0600 position and concentrated more on the fishing vessel. [2.2.2]

6. The bow of the heavily laden fishing vessel was damaged and pushed down in the collision, allowing seawater to enter the hull and flood progressively through the open starboard door at frame 74. [2.3]
7. With internal doors open, and especially the starboard door at frame 74, the fishing vessel had no effective subdivision and was unable to remain afloat after the hull had been opened to the sea. [2.3]
8. A positive conclusion cannot be made as to whether or not the chief officer left the bridge or whether this allegation was a contributory factor. [2.2.2]

3.4 OTHER FINDINGS

1. If the starboard door at frame 74 had been properly secured closed by the dogs, the vessel would have probably survived. [2.3]
2. Had the vessel been in the deepest condition approved by the stability book, the waterline after the collision would have been just below the sill of the starboard door at frame 74. The vessel's movement caused by the collision and in the seaway would have led to progressive flooding over the sill of the door opening. [2.3]
3. If *Silvery Sea* had been fitted with a collision bulkhead up to the shelter deck she would have survived. [2.3]